ETH zürich

D-INFK Lehre



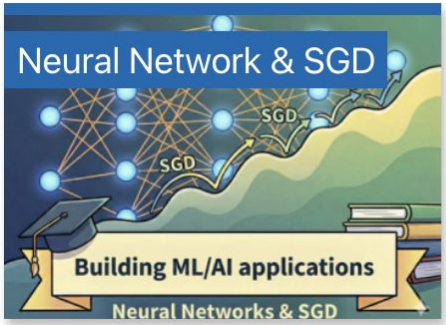
Project 4: Deepfake Generation & Detection

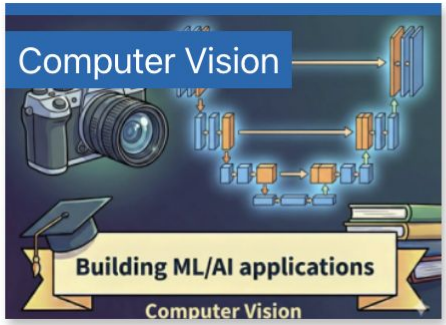
Carlos Cotrini

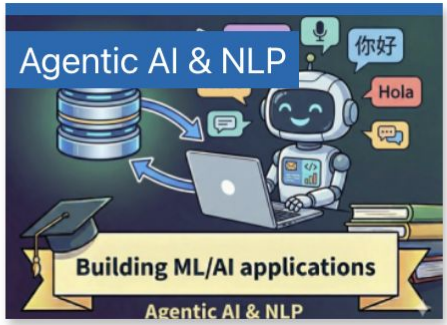
Christina Tsakanika, Daniele Russica

Building AI/ML Applications
2026

The Journey


Neural Network & SGD
SGD
SGD
Building ML/AI applications
Neural Networks & SGD


Computer Vision
Building ML/AI applications
Computer Vision


Agentic AI & NLP
你好
Hola
Building ML/AI applications
Agentic AI & NLP


Riemannian geometry &
GenAI
smiling
Building ML/AI applications
Riemannian geometry

Deepfake Generation & Detection

Image Inpaint Pipeline

Replace one fruit with another while preserving the background

Before



Original image with apple

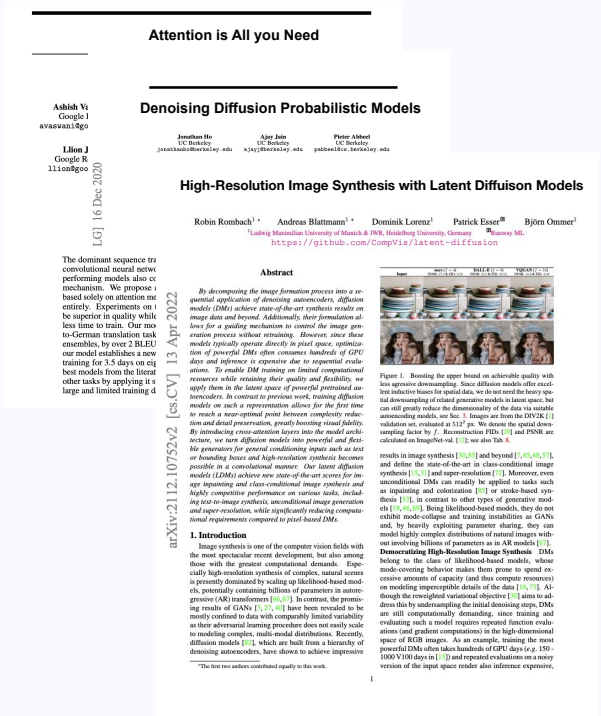
After



Generated image with orange

Reverse diffusion · T denoising steps · Conditioned on fruit embedding

Foundational Papers



Attention is All you Need

Denoising Diffusion Probabilistic Models

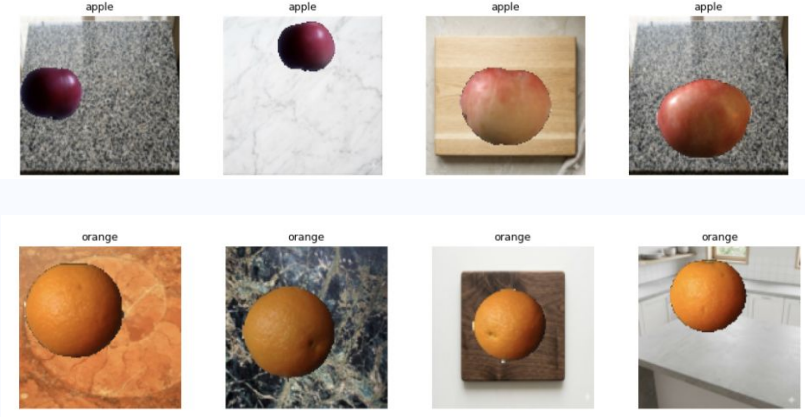
High-Resolution Image Synthesis with Latent Diffuison Models

Deepfake Generation & Detection

Goal

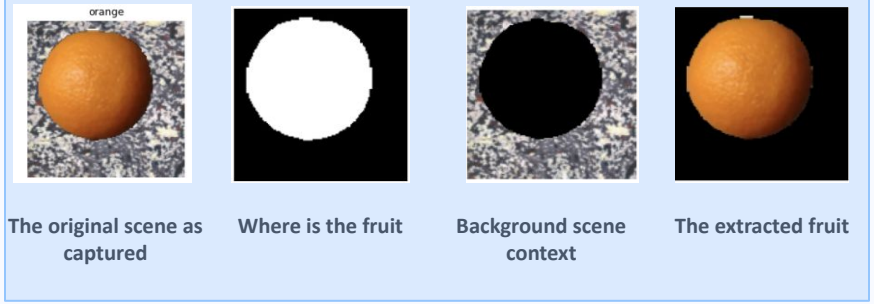
Train a U-Net to predict the noise pattern in corrupted images, conditioned on a fruit embedding
Run the reverse diffusion process to generate a new scene with the target fruit seamlessly in place

Dataset



4 versions per image

Composite	Mask	Redacted	Crop



The original scene as captured	Where is the fruit	Background scene context	The extracted fruit

Diffusion · Training

Phase 1 · Training

Input 1 Spatial context

Concatenated channel-wise before entering the U-Net

Noisy image

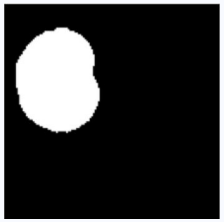


x_t (B,4,16,16)

corrupted at random timestep t

+

Mask



(B,1,16,16)

where is the fruit?

+

Redacted



(B,4,16,16)

background without fruit

cat (B, 9, 16, 16)

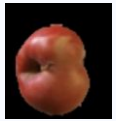
U-Net

pred noise e_theta

MSE loss = || ε - εθ ||^2

Input 2 Semantic conditioning

Which fruit should the model generate?



Fruit Encoder
CNN

Embedding (B, 128)

Crop (B,4,16,16)
the fruit alone

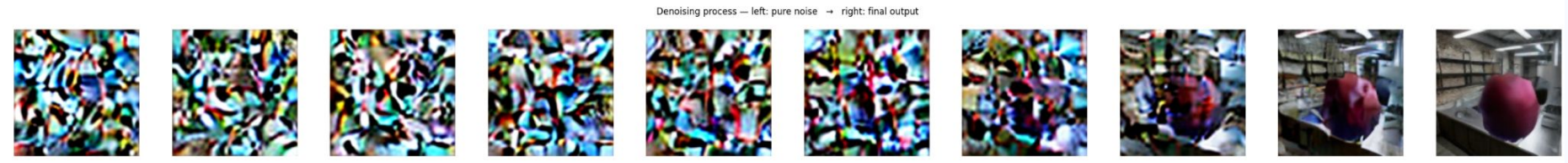
Cross-attention in U-Net

Each spatial position asks: 'what should I look like, given this fruit?'

Diffusion · Inference

Phase 2 · Inference

Start from random noise
Conditioning on a specific fruit embedding
Generate an image containing that fruit



Pipeline Overview

1 Data Preparation

composite, mask, redacted, crop

2 VAE Encoder

Pixel → latent space
composite, redacted

3 Fruit Encoder

crop → embedding

4 Diffusion U-Net

Noise prediction
cross-attention

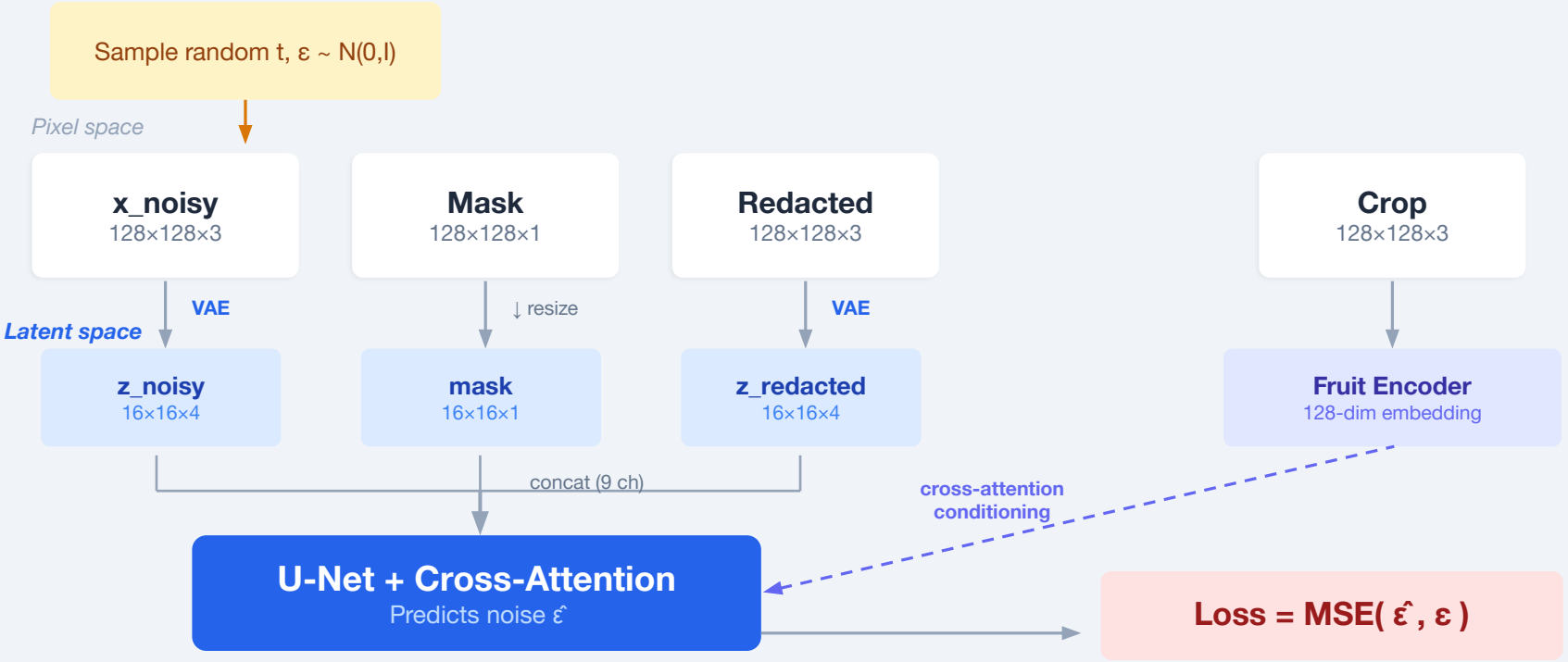
5 VAE Decoder

Latent → pixel space

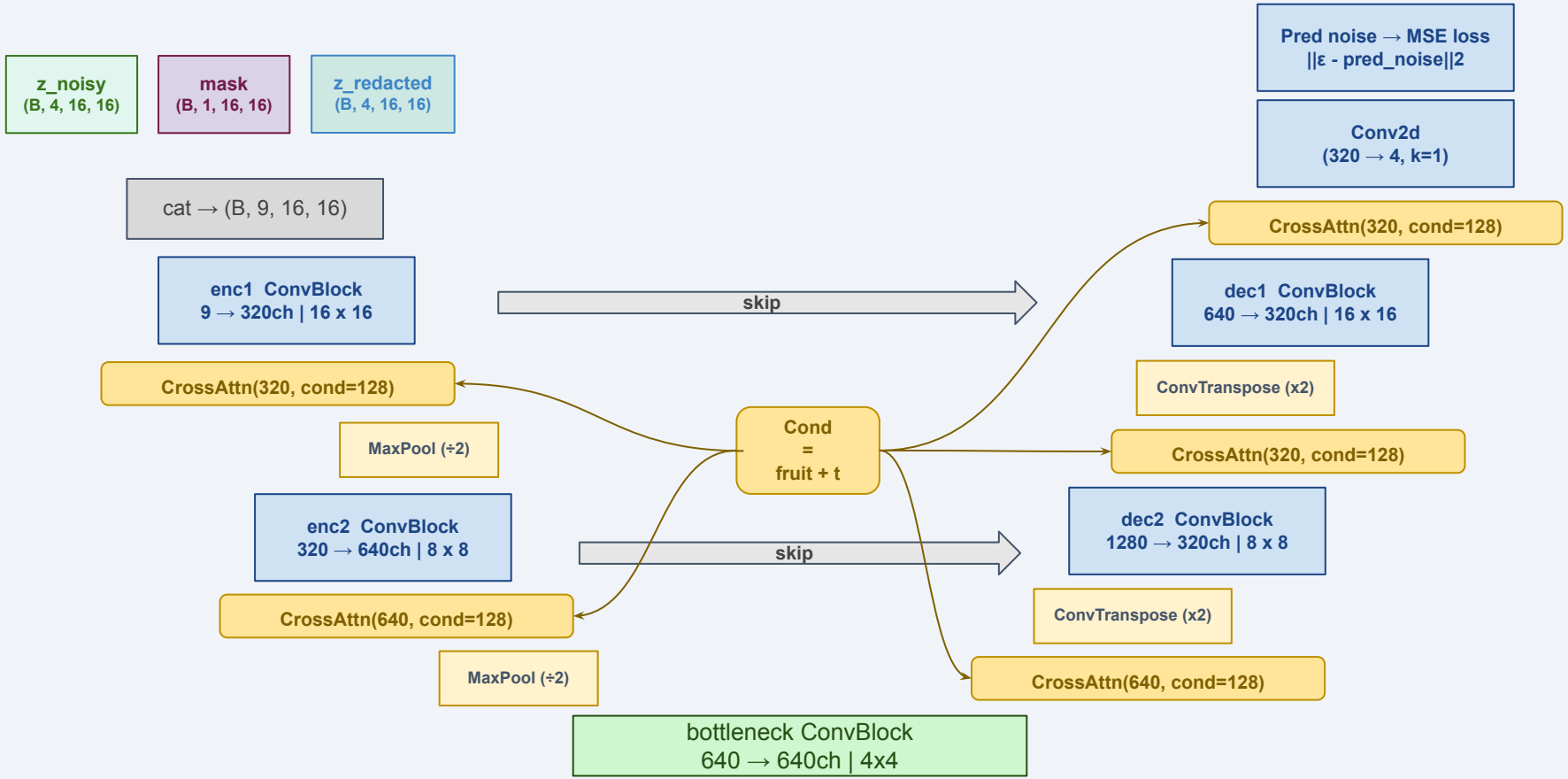
6 Discriminator

Real vs fake classification

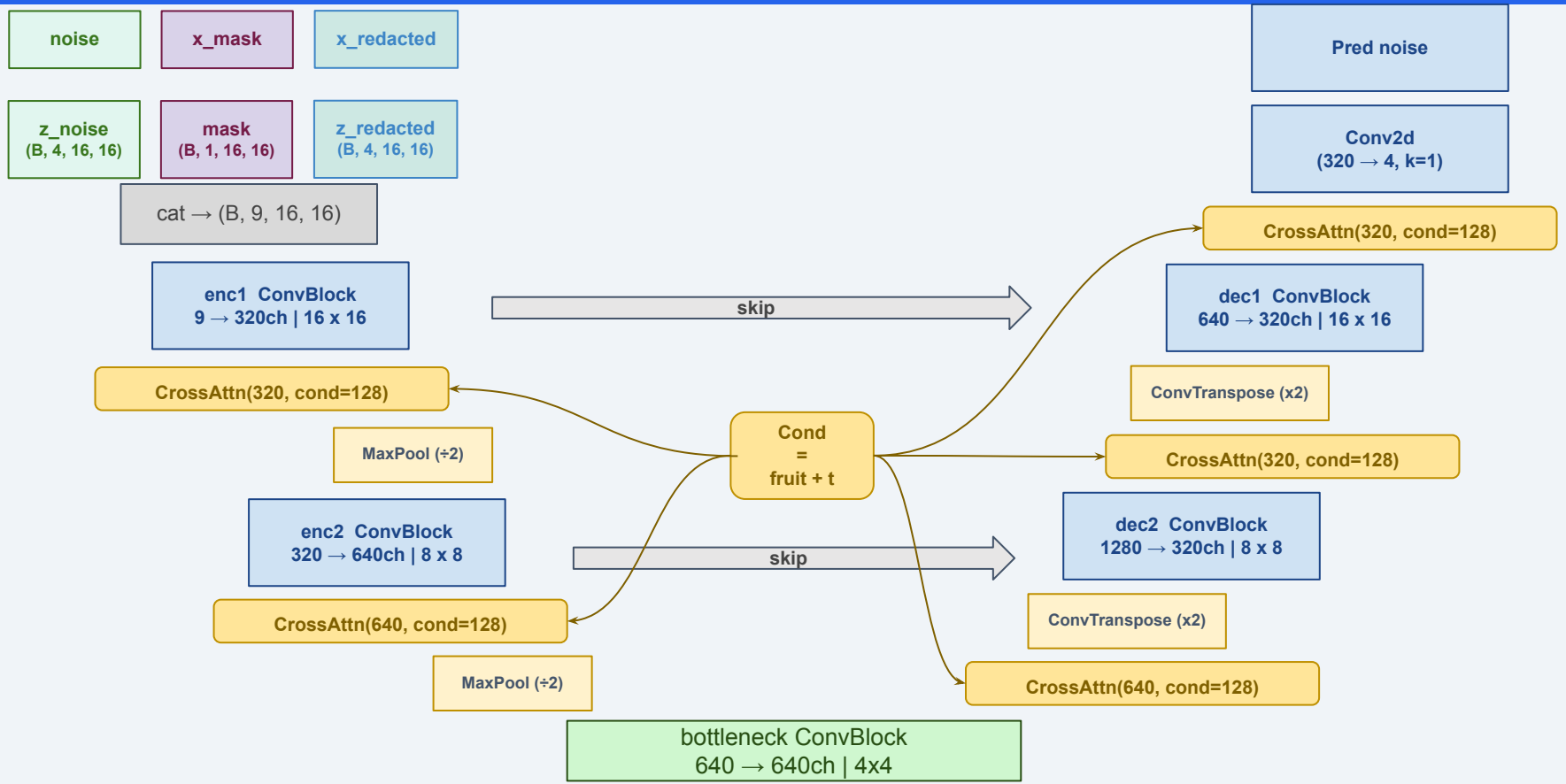
Training Process



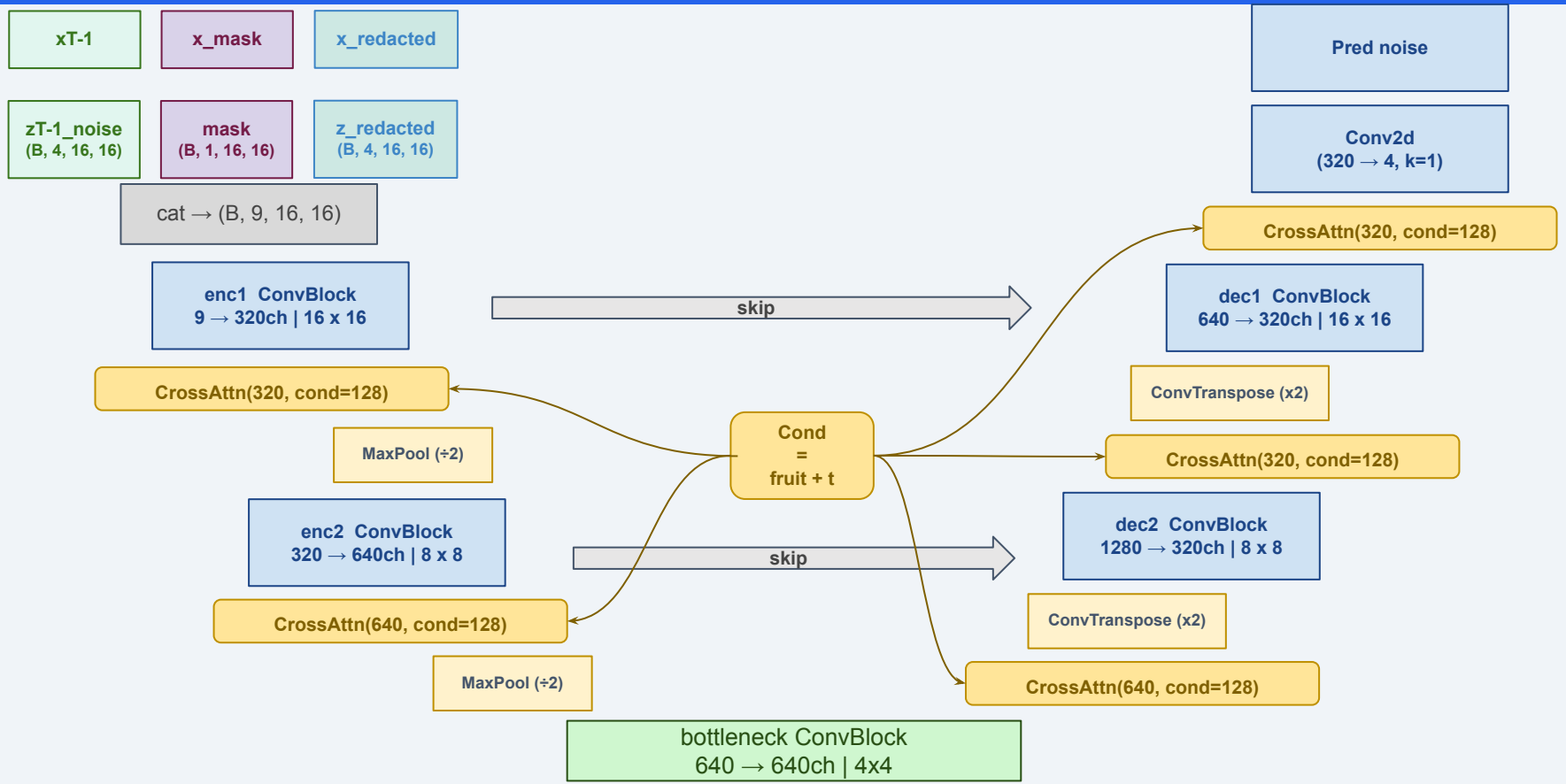
U-Net Training



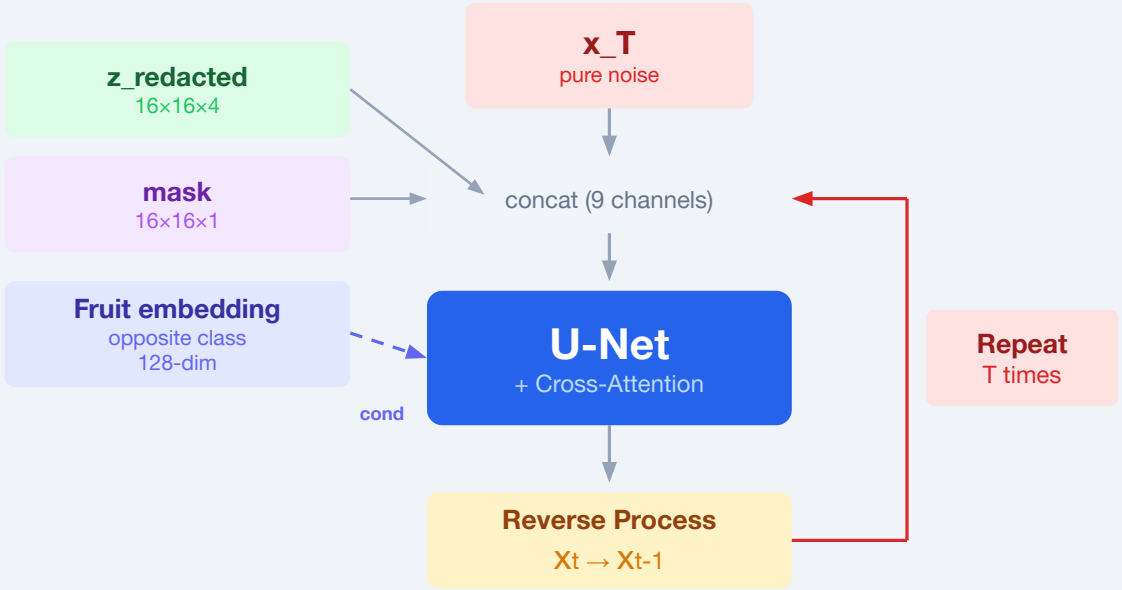
U-Net Inference



U-Net Inference



U-Net Inference



Denoising progression

t = T	Pure noise
t = 750	Faint structure
t = 500	Shapes visible
t = 250	Details emerge
t = 0	Clean image

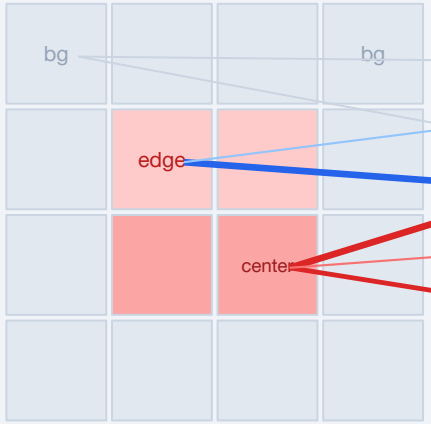
VAE Decoder
16×16×4 → 128×128×3

Deepfake image

Cross-Attention Mechanism

Feature map (Queries)

Each position asks: "What should I look like?"



Fruit embedding (Keys & Values)

4 synthetic tokens projected from 128-dim embedding

Token 1
Color info

Token 2
Shape info

Token 3
Texture info

Token 4
Size info

Center pixel
"What color and texture should I have?"

Edge pixel
"What shape contour should I follow?"

Background pixel
Weak attention
fruit info not needed

Line thickness = attention weight

strong — weak